Computer Networks Lab
Fall 2025
Week 05

Socket Programming

<u>Lab Task</u>

<u>Task 1:</u>

Build a TCP client-server application to check whether a string is a palindrome.

Server must:

- Receive a string from the client.
- Check if the string is a palindrome (same forwards and backwards, e.g., “level”, “madam”).
- Send back a message: either “Palindrome” or “Not a palindrome”.

Client must:

- Prompt the user for input, send it to the server, and display the result.
- Communication continues until the client sends “quit”, at which point the server closes the connection.

<u>Example:</u>

Client sends: “civic” → Server replies: “Palindrome”
Client sends: “network” → Server replies: “Not a palindrome”

<u>Task 2:</u>

Build a TCP server and client to implement a simple calculator that can perform basic arithmetic operations. The server will handle the computation, while the client will gather input from the user and display the result.

The server must:

- Receive two operands (numbers) and the operation choice from the client.

- Perform the requested operation (**Addition, Subtraction, Multiplication, Division**).
- Handle special cases such as:
    - Division by zero (send an error message back to the client).
    - Negative numbers (ensure they are processed correctly).
- Send the computed result or error message back to the client.

The client must:

- Prompt the user for two numbers and an operation choice.
- Transmit this information to the server.
- Display the result (or error message) received from the server.

Communication continues until the client chooses to **exit**. The server should close the connection gracefully once the client terminates.